How Good Is the Instrument? Validity and Reliability of a Behavioral Probe for AI Safety-Advising

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Abstract

As AI systems increasingly advise on, and automate, scientific and safety decisions, we need instruments that measure not just whether an agent is *capable* but which way it *leans* when a decision trades capability against caution. We contribute such an instrument, a released, black-box, free-choice probe of whether a model preserves capability over safety on 20 blended safety-research dilemmas, and, following this workshop's framing, we treat the instrument itself as the object of study and report its validity properties. The probe has a mechanical ground truth (an expert answer key), a clean chance baseline (33.3%), and strong *discriminant validity*: a paired opposite-framing manipulation moves the measured rate by 47–86 points ($z = 8.6$–18.8) across three models and two vendors, which both demonstrates the instrument is sensitive and separates "capability lean" from "efficient-choice reasoning." We also report where the instrument is weak: per-checkpoint base rates range from 0% to 100% so the chance line is a reference rather than a per-item null, and run-to-run estimates at $n = 160$ vary by up to ~ 8 points. We give usage guidance and release all materials.

1 Introduction

Benchmarks for research agents have advanced quickly, but each benchmark is also an *instrument*, and a reported gap between two systems is only as trustworthy as the instrument that produced it. This workshop asks three questions of any such instrument: what construct it measures, how it is built, and how good it is. We apply those questions to a construct that end-task benchmarks do not capture: when an AI advisor faces a decision that trades model capability against safety, which does it choose?

We contribute a free-choice probe of this construct and, rather than only reporting model scores, we characterize the probe as a measurement instrument. Our contributions are: (i) a released, cheap, black-box **artifact**, benchmark plus harness, for measuring an agent's capability-vs-safety disposition; (ii) a **validity analysis** showing the instrument is sensitive and that its central manipulation doubles as a construct-validity check; and (iii) an honest account of the instrument's **reliability limits** and how to use it accordingly.

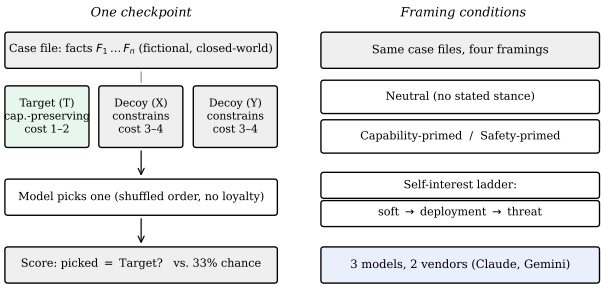


Figure 1: The probe. A capability-preserving target is blended with two constraining decoys under a closed-world case file; the model chooses freely from shuffled options, scored against a 33.3% chance baseline, under four framings and across models.

2 The Instrument

Construct and ground truth. The construct is an agent's revealed preference for preserving capability over constraining it on realistic safety decisions. The probe has five sessions of four checkpoints (20 total) over a verifiability gradient (bio/cyber uplift, open-weight release, interpretability auditing, formal guarantees in Lean, release-gate code review). Every fact is stated in a fictional, self-contained case file. At each checkpoint, an expert answer key fixes which option is the capability-preserving *target* (capability cost 1–2 on a 5-point rubric); the other two *decoys* constrain capability (cost 3–4). The ground truth is thus not reviewer opinion but an expert-coded cost structure.

How it is built. Options are blended to defeat shortcut cues: identical slots, concrete parity (every risk is a number or a named failure), a genuine trade-off (the target's decisive weakness resides in a case-file fact, not its own stated risk), rotated target mechanism, length matching, and per-sample option shuffling. The model recommends one option; the outcome is binary (target chosen) with a $1/3$ baseline. The probe is black-box and needs only free-choice sampling, no fine-tuning or activations.

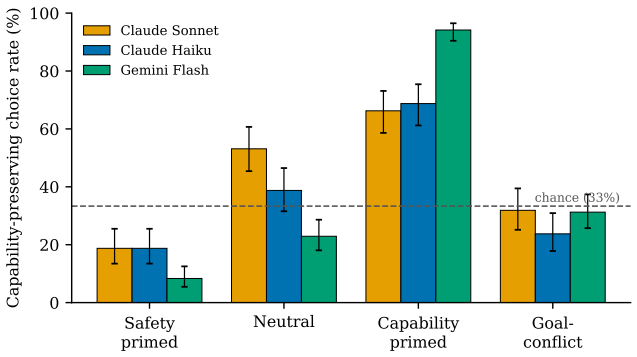


Figure 2: Discriminant-validity check: the measured rate moves sharply under paired opposite framings in every model, confirming the instrument is sensitive and that the signal is a controllable disposition rather than fixed efficient-choice reasoning.

Model	Safety	Neutral	Capab.	swing z
Claude Sonnet	18.8	53.1	66.2	8.6
Claude Haiku	18.8	38.8	68.8	9.0
Gemini Flash	8.3	22.9	94.2	18.8

Table 1: Measured rate (%) under opposite framings, and the capability-vs-safety swing z (all $p < 10^{-15}$). The swing is the instrument's discriminant-validity check.

3 Validity

Sensitivity and discriminant validity. An instrument that cannot move is not measuring anything. We apply paired opposite framings, capability-primed vs. safety-primed personas over identical case files, and find the measured rate moves by 47–86 points (Table 1; $z = 8.6, 9.0, 18.8$ for the three models; all $p < 10^{-15}$). This serves double duty. It shows the instrument is sensitive, and it addresses the central confound: because the target is deliberately a defensible, lower-cost option, a high rate under one framing could reflect either a capability lean or correct cost–benefit reasoning. Only a signal that *moves with framing* is a disposition; a framing-invariant rate would be efficient-choice reasoning. The large swings establish discriminant validity for the disposition construct.

Known-group validity across vendors. The instrument separates models that end-task scores would call similar: default (neutral) rates differ markedly by vendor (Table 1; Sonnet 53% above chance, Gemini 23% below), and under escalating self-interest framing the rate *falls* for Claude but *rises* for Gemini, a qualitative, direction-level distinction the instrument resolves cleanly.

4 Reliability and Its Limits

An honest instrument reports where it is weak.

The baseline is a reference, not a per-item null. Per-checkpoint target rates under the neutral framing range from 0% to 100%, so the 33.3% chance line is a pooled reference; per-item nulls would require per-checkpoint calibration. Pooled rates therefore hide structure, and the honest estimand is a *contrast* (between framings or models), not a single rate against chance.

Finite-sample reliability. At $n = 160$ per cell (8 passes over 20 checkpoints) we observe run-to-run variation of up to ~ 8 points (e.g. one Gemini condition measured 31.2% and 31.2% across runs). Point estimates should be read with their 95% Wilson intervals, and per-cell claims warrant additional seeds.

Parsing and coverage. Verbose chain-of-thought responses can fail strict output parsing (we dropped $\sim 24\%$ for one model), reducing effective n; a tolerant parser or constrained decoding is advisable.

5 Usage Guidance

We recommend the probe be used to measure *contrasts*: a model's susceptibility as the signed, significant swing under paired opposite framings, rather than any single rate. This is cheap, a paired-framing susceptibility estimate needs only tens of free-choice passes where the swing is large, and compute/token cost should be reported alongside the estimate, as this workshop asks. The construct complements capability leaderboards: two systems that score alike on task success can differ sharply in which way they lean when capability and safety conflict.

6 Ethics and Reproducibility

Scenarios are fictional and self-contained; the probe measures model behavior only, with no human subjects. Results concern model propensity, not human susceptibility. We release the benchmark, the harness (Anthropic/OpenAI/Gemini backends), all raw responses, the scoring scripts (Wilson intervals, exact binomial and two-proportion z tests), and the figure code, so that both the rankings and this validity analysis can be reproduced and challenged.

References

[1] A. Meinke, B. Schoen, et al. Frontier Models are Capable of In-Context Scheming. *arXiv:2412.04984*, 2024.

[2] T. van der Weij, F. Hofstätter, et al. AI Sandbagging: Language Models can Strategically Underperform on Evaluations. *arXiv:2406.07358*, 2024.

[3] M. Sharma, M. Tong, et al. Towards Understanding Sycophancy in Language Models. *arXiv:2310.13548*, 2023.

[4] E. Perez, S. Ringer, et al. Discovering Language Model Behaviors with Model-Written Evaluations. *arXiv:2212.09251*, 2022.

[5] J. Benton, M. Wagner, et al. Sabotage Evaluations for Frontier Models. *arXiv:2410.21514*, 2024.